Newton's Laws with a Pulley Cart

U3 / S3.1

PREP — BEFORE DAY 1 AND DAY 2

WHEN	TASK	DETAIL
Days before	Charge and number	Charge 5 sensor carts and 5 GDX-FORs. Every group has one of each (confirmed by Matt, 2026-11-12). Number carts and sensors 1–5 so groups connect to their own pair.
Days before	Check signs	Roll one cart toward the pulley and pull its hook. Note whether velocity and force read positive. The handout has students record F and a as positive numbers in Part B, so a sign mismatch can't make the slope negative. If velocity reads negative toward the pulley, the Part A slope comes out small and *positive*. It means the same thing.
Days before	Test Part C yourself	**Matt to confirm:** (1) where the string attaches on the back of the GDX-FOR (eyelet, tie point or strap slot); (2) whether the GDX-FOR droops when it hangs between the cart hook and the string. Its weight is comparable to the 0.5–1 N pull. A drooping sensor reads along a tilted axis, and the two readings drift apart. If it rests on the track instead, its sliding friction may be too large for 50 g to move it. See the backups below.
Before Day 1	Trays and tracks	5 tracks, a pulley clamped to one end, an end stop on the other. Tray per group: cart, mass bar, about 1.5 m of string, scissors, meter stick, hanger and slotted masses, folded towel. Balances within reach. **Run one Part A trial yourself** with 20 g and check the flat stretch is long enough to fit.
Before Day 2	GDX-FOR	One GDX-FOR per group, hook screwed in. Strings stay tied from Day 1 if the setups stay up.

Materials — total for 5 groups

Go Direct Sensor Cart × 5 · GDX-FOR × 5 (Day 2) · Dynamics Track × 5 · clamp-on pulley × 5 · end stop × 5 · cart mass bar × 5 · mass hanger and slotted masses, 20–100 g per group, × 5 · string, about 1.5 m × 5 plus spares · folded towel × 5 · meter stick and scissors × 5 · balance · student devices with Graphical Analysis × 5. **Matt to confirm:** the hanger's own mass. The handout uses 20, 40, 60 and 80 g with the hanger included. If the hanger alone is 50 g, use 50, 60, 70 and 80 g (empty cart) and 50, 70 and 90 g (cart + bar). Track length is not confirmed either. The "start about 1 m from the pulley" step assumes a 1.2 m track or longer.

TIMING — TWO 50-MINUTE PERIODS (RECOMMENDED)

MIN	PHASE	WATCH FOR
D1 0–10	Pre-lab check, setup: pulley, level, string, loops	Strings cut to the wrong length. Pulley not adjusted to make the string level.
D1 10–20	Part A: 3 trials	Fitting across the speeding-up part. Pushing the cart on release.
D1 20–45	Part B: 4 runs empty, 3 runs with bar, F–a graph	Averages taken across the slack part. Bar added without re-massing.
D1 45–50	Reset	Masses back in the box. Setups can stay up for Day 2.
D2 0–5	Hook up the GDX-FOR, connect, zero both	Sensors zeroed with the hanger still pulling.
D2 5–20	Part C: 50 g and 100 g, held then moving	GDX-FOR drooping or twisted out of line.
D2 20–45	Post-lab in notebooks	Units on every number.
D2 45–50	Reset	

Parts A–C plus setup run about 45 minutes of data taking, before any analysis. That does not fit one period with the five-minute reset, so keep the two days Matt approved. Day 2 has room for the whole post-lab in class.

FAILURE POINTS

- **String not level.** If the string angles up from the hook to the pulley, part of the pull is vertical and F reads high against a. Raise or lower the pulley (Setup step 6).
- **Hanging mass swinging.** A swinging hanger makes the force reading wobble. Let it hang still before releasing the cart.
- **Not zeroing with nothing pulling.** Zero with the hanger lifted, before every run. In Part C, zero both sensors that way. A leftover offset shifts the Part B intercept and the Part C comparison.
- **The string must go slack cleanly in Part A.** If the string is too long, the cart reaches the pulley before the mass lands and there is no flat part. If the mass bounces off a hard floor, the string snaps tight again and puts a spike in the flat part. That's what the towel and the Setup step 5 check are for.
- **Fit and average regions.** Part A fits only the flat part. Part B averages only the steady speeding-up stretch, leaving out the release spike and everything after the slack.
- **Catching late.** With 80 g the cart reaches about 1.6 m/s on 0.6 m of travel. Someone's hand stays near the pulley.
- **Part C sign.** Both sensors read a pull, often with the same sign. That is correct: each sensor measures the pull on itself, and the two face each other. Students compare sizes, with no sign flipping.

Backups for Part C if the GDX-FOR droops

(1) Ride the GDX-FOR on a spare cart, if there are enough spares: the sensor stays level and adds little friction. (2) Matt's original idea: hang the GDX-FOR in the vertical part of the string, between the string and the mass hanger. Both sensors then read the same string tension. That shows the pull is passed along the string, but it is *not* a single action–reaction pair, so the Part C wording on the handout would need rewriting.

STRING-LENGTH ESTIMATE — CHECK

The string is a horizontal piece (hook to pulley) plus a hanging piece (pulley to hanger). The hanger lands when the hanging piece is about the pulley height H. Whatever is left is the horizontal piece, which is how far the cart still is from the pulley. So for 40 cm of coasting room, $L \approx H + 0.40$ m, wherever the cart starts. Example: H = 0.95 m gives L = 1.35 m. Starting 1.00 m out, the hanger starts 0.35 m below the pulley and falls 0.60 m. The cart speeds up for 0.60 m and then has 0.40 m to coast. The hanger's own height only adds coasting room, so the check in Setup step 5 says "at least 40 cm." **A tempting version, $L \approx d + H - 0.40$ m, is wrong:** it makes the mass fall only 40 cm and leaves the cart d − 0.40 m from the pulley.

EXPECTED RESULTS — SAMPLE DATA

Sample data, not real measurements. These come from a model of the setup with a cart of about 0.28 kg and rolling friction of about 0.03 N, with realistic scatter added. The handout's example rows are rows from this set. Replace the masses with your own balance readings.

QUANTITY	ACCEPTED	SAMPLE	TYPICAL RANGE
Part A slope of flat part	0 with no friction	−0.094, −0.108, −0.097 m/s²	about −0.05 to −0.15 m/s²
Part B slope, cart only	0.282 kg (sample balance)	0.292 kg · 3.5%	within about 10%
Part B slope, cart + bar	0.779 kg (sample balance)	0.744 kg · 4.5%	within about 10%
Part C percent difference	0%	1.0% to 3.4%	under about 5% when lined up

WORKED SAMPLE ANALYSIS — SAMPLE DATA, NOT REAL MEASUREMENTS

Sample cart 0.282 kg, sample bar 0.497 kg, so cart + bar = 0.779 kg. g = 9.8 m/s².

Part A — first law (20 g)

TRIAL	HANGING MASS (g)	V AT START OF FLAT (m/s)	V AT END OF FLAT (m/s)	SLOPE (m/s²)
1	20	0.81	0.78	−0.094
2	20	0.83	0.79	−0.108
3	20	0.80	0.77	−0.097

mean slope = $\frac{-0.094 + (-0.108) + (-0.097)}{3} = \frac{-0.299\ \text{m/s}^2}{3} = -0.0997\ \text{m/s}^2 \approx -0.100\ \text{m/s}^2$

For you, not asked on the handout: friction f = m|a| = (0.282 kg)(0.0997 m/s²) = 0.028 N. That is close to the Part B cart-only intercept, 0.023 N.

Part B — second law

RUN	CART SETUP	HANGING MASS (g)	F (N)	a (m/s²)
1	cart only	20	0.18	0.53
2	cart only	40	0.33	1.06
3	cart only	60	0.48	1.57
4	cart only	80	0.60	1.97
5	cart + bar	40	0.37	0.39
6	cart + bar	60	0.53	0.61
7	cart + bar	80	0.69	0.82

Least-squares fits: cart only, slope 0.292 kg, intercept +0.023 N. Cart + bar, slope 0.744 kg, intercept +0.079 N.

Cart only: percent error = $\frac{|0.292\ \text{kg} - 0.282\ \text{kg}|}{0.282\ \text{kg}} \times 100 = \frac{0.010\ \text{kg}}{0.282\ \text{kg}} \times 100 =$ **3.5%**

Cart + bar: percent error = $\frac{|0.744\ \text{kg} - 0.779\ \text{kg}|}{0.779\ \text{kg}} \times 100 = \frac{0.035\ \text{kg}}{0.779\ \text{kg}} \times 100 =$ **4.5%**

Slope rise: 0.744 kg − 0.292 kg = 0.452 kg, against 0.497 kg added (9% low). The intercept is bigger with the bar on because a heavier cart has more rolling friction.

Part C — third law

RUN	HANGING MASS (g)	HELD OR MOVING	CART SENSOR (N)	GDX-FOR (N)	PERCENT DIFFERENCE
1	50	held	0.49	0.48	2.1%
2	50	moving	0.30	0.29	3.4%
3	100	held	0.98	0.97	1.0%
4	100	moving	0.53	0.54	1.9%

Run 1: average = $\frac{0.49\ \text{N} + 0.48\ \text{N}}{2} = 0.485\ \text{N}$ percent difference = $\frac{|0.49\ \text{N} - 0.48\ \text{N}|}{0.485\ \text{N}} \times 100 = \frac{0.01\ \text{N}}{0.485\ \text{N}} \times 100 =$ **2.1%**

Run 2: $\frac{0.01\ \text{N}}{0.295\ \text{N}} \times 100 =$ **3.4%** Run 3: $\frac{0.01\ \text{N}}{0.975\ \text{N}} \times 100 =$ **1.0%** Run 4: $\frac{0.01\ \text{N}}{0.535\ \text{N}} \times 100 =$ **1.9%**

Held check: W = mg = (0.050 kg)(9.8 m/s²) = 0.49 N and (0.100 kg)(9.8 m/s²) = 0.98 N, matching the held readings.

Moving readings are smaller. The pull between the sensors only has to accelerate the cart and beat its friction: (0.30 N – 0.027 N) ÷ 0.282 kg gives a ≈ 0.97 m/s² for 50 g, and about 1.8 m/s² for 100 g. The GDX-FOR's accelerometer should show about the same value.

PRE-LAB KEY

1. Newton's three laws. (1) An object stays at rest or keeps moving at a constant velocity unless a net force acts on it. (2) ∑F = ma: net force causes acceleration; more force, more acceleration; more mass, less acceleration. (3) When A exerts a force on B, B exerts a force of the same size in the opposite direction on A. Full credit for each law in the student's own words with its key idea.
2. String length with H = 0.95 m. 0.95 m + 0.40 m = 1.35 m, loop to loop.
3. Predicted v–t graph. A straight line rising from zero (speeding up at a steady rate), then a nearly flat line starting where the string goes slack, sloping down very slightly, then a sharp drop at the catch. Credit needs the rise, the flat part and the slack label.

POST-LAB KEY — FROM THE SAMPLE DATA

1. First law, Part A. Once the mass landed, the string went slack and nothing pulled the cart forward, so the net force was about zero. The velocity stayed nearly constant (0.81 to 0.78 m/s), as the first law says. It was not perfectly flat: the slope was about –0.10 m/s² because rolling friction still acted backward. No credit for "the force from the string kept it going." That is the misconception this part exists to fix.
2. Second law, Part B. As F went up, a went up in proportion: the points make a straight line. With the bar on, the same F gave a smaller a, so the line was steeper. Its slope, the mass, went from 0.292 to 0.744 kg.
3. Percent error. 3.5% (cart only) and 4.5% (cart + bar). Work is above.
4. Error source and direction. Needs a specific cause and its direction. Good answers: the string angled up from the hook, so the sensor read the whole pull but only the level part sped the cart up: F too big for its a, slope (mass) too *high*. Or the averaged stretch ran into the slack part or the release spike, which drags points off the line in either direction. Or the hanger was swinging, which makes F noisy. Watch for "forgot to zero": that shifts the intercept, not the slope, so it does not explain a wrong mass. Partial credit only. "Human error" earns nothing.
5. Third law, Part C. The two readings were the same size at every moment (within about 0.01 N), held or moving. The GDX-FOR pulled the cart toward the pulley, and the cart pulled the GDX-FOR back toward the cart: equal size, opposite direction, acting on different objects.
6. Percent difference. 2.1%, 3.4%, 1.0%, 1.9%; work above.

TEMPLATE EXCEPTIONS

This lab departs from the locked lab template: no Safety section, no Disposal/Alconox block, filled example tables, and a reduced pre-lab and post-lab. These exceptions are approved and recorded in SHULL-CHG-0073.

GRADING

No point split for Physics labs is set yet. Matt sets this. The carbonless copy is graded individually.

MISSED LAB

A student who misses a day completes a Missed Lab Analysis rather than re-running the lab. Give them one group's Tables 1–3 from the same period and a screenshot of that group's Part A v–t graph and F–a graph. They answer all six post-lab questions from that data.